

K - Reflection of the design project

This project incorporated many different types of components to be completed and most importantly, addressed the problem given by our primary stakeholder, Ms. Tatro, the library of ERHS and its school community.

We were considering possible designs. One of which was a traditional bookend that has an L shape. It is meant to hold a book from the spine up. It could also hold a book while it's open. This is an improvement.

Our prototype was finally decided as a wooden structure with a rectangular base in which the bottom edge/tail of the book would rest on and a triangular wooden structure in the middle of the rectangular base for the books to lean on, causing the orientation of the books to be upright (spine pointing upwards.) Following our plans, we cut the wood and binded the wooden pieces together with wood glue. We also incorporated four rubber grips at the bottom of the rectangular base to address friction issues and the deterioration of the main component.

The tools you have accessible are also important. Having high quality sandpaper for the wood, different glues for the wood and rubber grips, and a corner/right angle clamp for letting the wood glue dry in place can be very helpful and help save time

The project met our durability and structural integrity requirements, being able to withstand 5 kg of force on one side and remain intact without the structure bending (though it is only recommended to add 3.5k per side.) The structure can also easily be moved around.

The structure also met our sustainability goals. All materials were either recycled (the wood), repurposed (the paint), or donated by Mr. David Eisenberg, Mr. Robert Nusum, and Mr. Lowie Paz.

Though our project was a success, there were many compromises we had to make throughout the creation part of the project, ones that were not fully realized until the building of the prototype. For one, the wooden structure used for the project was sturdy but was larger and bulkier than intended which negatively affects shelving space. It is important when conducting future projects that time is spent looking for the best materials especially if they are recycled.